Mathematical destination

A state records **both** the walker's position and the lamps that are on. Each move costs one: walk one place left or right, or flip just the lamp under the walker. Starting with all lamps off at 0, the minimum number of moves is

number of target lamps on + shortest walk visiting those lamps and ending at the target position.

Every target lamp needs an odd number of flips, hence at least one. Extra flips cannot shorten the necessary walk. Conversely, a shortest visiting walk can flip each required lamp exactly once. This proves the decomposition, rather than merely suggesting it from trials.

The destination is a **dead end for distance from the start**: the state with lamps $-1, 0, 1$ on and walker at 0 has distance 7, yet each of its three one-move neighbors has distance 6. A longer move word can therefore describe an easier target. "Dead end" does not mean the walker is unable to move.

Which targets are dead ends. Exactly those with the walker at 0, lamp 0 on and at least one lamp on each side of 0. Write $\ell < 0 < r$ for the end lamps. With the walker between them the walk is $2(r - \ell) - |p|$, so at 0 a step either way shortens it by one, and turning lamp 0 off saves a flip without moving an end. For any other target some move costs one more: a step toward 0, a step past an end lamp, a step toward a side with no lamp, or switching lamp 0 on. One move always changes the distance by exactly one (lit lamps plus walker position changes parity each move), so "easier" and "not harder" agree. Examples: Problem 3's target (7), lamps $-2, 0, 1$ or $-1, 0, 2$ (9), Problem 2's B (11).

Assumptions and scope. The mathematical street is the integer line with finitely many lamps on. All printed targets have optimal routes inside the displayed -4 to 4 street; no edge extension is needed for this packet. Walks and flips are separate moves. Distance always means fewest moves *from all-off at 0*, not from the preceding target.

What children discover. Experiment with move words; conjecture which routes are shortest; then justify them by unavoidable flips and walking. Grades 3–5 share the concrete entry. Prerequisites are reading signed positions, counting a short word, and preserving a two-part state. An adult can read and record. General formulas are an optional older-table continuation, not an entry requirement.

Materials and preparation

Per pair: nine two-sided lamp counters, one visibly different walker pawn, the five student pages, and a pencil. The large lamp circles are 13 mm across on 19 mm centres at actual size, so use counters 17 mm across or smaller, or paper squares dark on one side; common 2–2.5 cm counters overlap their neighbours and hide which lamp is lit. Keep each counter on its own lamp, since one moved beside a lamp loses its position, and stand the walker just above its lamp, where the home mark is at 0, not on the counter. Counter fit is a prediction until rehearsed. Print single-sided at 100%; no cutting or specialist kit is needed.

K–1 tables. There is no K–1 packet. An adult-led route on these pages, unpiloted: the launch, then Problem 1's three targets and Problem 2's A and C as picture matching. Children build the picture with counters and the pawn while the adult reads and writes the word; each move adds one cube to a stack, and stacks are compared for "fewest". *Key in plain words:* 1A, step right and flip, 2 moves; 1B, step left, flip, step right twice, flip, 5; 1C, flip, step right twice, flip, step back twice, 6; 2A and 2C, 9 each. About a dozen cubes per child.

For the two older tables, prepare three working sets: two pairs of third graders, and an older pair with a rotating referee. One adult anchors each table. A partner controls flips and first points to the walker's location; the other controls movement. Swap roles between targets. Keep one move word as the recoverable record; count it afterward.

Brief common launch and first route

Let pairs handle the counters. Use the printed non-target demonstration: start at -2 with lamps off, execute `R`, then `F`; the walker stays at -1 during the flip. Have a partner replay `RF` and name both parts of the resulting state. Then show the goal with a non-task target, lamp -1 on and the walker at 0: replay `LFLRR` (5 moves) and ask whether a shorter word reaches it (`LFR`, 3). Reset all-off at 0 for Problem 1.

Allow roughly 10–15 minutes for Problem 1 and 10–15 for Problem 2. Pause when children can replay and improve a route. A feasible first visit ends with Problem 3 and a shared lower-bound explanation; reserve Problems 4–6 for remaining time or a return visit. The older table can reach the dead-end comparison sooner. These times are planning estimates, not classroom observations.

Numbered answers and optional hints

All words below are read left to right and start all-off at 0. Different shortest words are welcome. The walker position is p; S is the set of lit lamps.

Problem 1. The three targets have the following minima.

Target	(p, S)	One shortest word	Walks + flips	Total
A	$(1, \{1\})$	`RF`	$1+1$	2
B	$(1, \{-1, 1\})$	`LFRRF`	$3+2$	5
C	$(0, \{0, 2\})$	`FRRFLL`	$4+2$	6

For B, visiting -1 then ending at 1 needs at least three walks. For C, reaching 2 and returning needs at least four. *Hint:* temporarily ignore flips and trace only where the walker must go.

Problem 2. All three have lamps $-2, 0, 2$ on, so three flips are unavoidable. A ends at -2: **9**, using `FRRFLLLLF`. B ends at 0: **11**, using `FLLFRRRRFLL`. C ends at 2: **9**, using `FLLFRRRRF`. To end at an extreme requires at least six walks: two to the other extreme, then four across. Returning to 0 requires eight. *Hint:* compare only the final walker positions; the lit lamps have not changed.

Problem 3. 7, for example `FLFRRFL`. Three target lamps force three flips. Visiting both -1 and 1 and returning to 0 forces four walks. The exhibited word attains $3+4$. A trial with seven moves alone is only an upper bound; the unavoidable-moves argument proves optimality.

Problem 4. Reset to Problem 3 before each of the following changes. The minima in the last column are from the *original* start.

Change	New (p, S)	Shortest word from start	Walks + flips	Total
L	$(-1, \{-1, 0, 1\})$	`FRFLLF`	$3+3$	6
R	$(1, \{-1, 0, 1\})$	`FLFRRF`	$3+3$	6
F	$(0, \{-1, 1\})$	`LFRRFL`	$4+2$	6

The same required-visit and flip arguments give all three lower bounds. Do not merely append a letter to the seven-move word: that produces an eight-move route, not a shortest route. *Hint:* build the new target, reset completely, and solve it afresh.

Problem 5. No. The Problem 3 target has no next move that makes it harder: Problem 4 checked *every* legal next move, and L, R and F all reduce the distance from 7 to 6. One unsuccessful attempt would not settle an "every target" question; these are the complete alternatives at that state.

Problem 6. Exactly the targets with the walker at 0, lamp 0 on and at least one lamp on each side (see the dead-end paragraph on page 1). Problem 2's B is one (11; L, R and F each give 10). The smallest others are lamps $-2, 0, 1$ and $-1, 0, 2$ (9; each move gives 8), then $-2, -1, 0, 1$ and $-1, 0, 1, 2$ (10). Worth testing as non-examples: lamps -1 and 1 with the walker at 0 (6), where F gives 7, and Problem 1's C (6), where L gives 7. *Hints, in order:* change one thing about the Problem 3 target and test all three moves; ask what each move does to the walking and to the flips; ask which lamps must stay on.

General rule and limits

For an optional extension, let $\ell = \min(S \cup \{0\})$ and $r = \max(S \cup \{0\})$. The exact distance to (p, S) is

$$|S| + \min\bigl(-\ell + (r - \ell) + |p - r|, \quad r + (r - \ell) + |p - \ell|\bigr).$$

A walk must visit the two extremes in one order or the other. Each displayed length is necessary for its order and attained by walking directly between those stops, flipping required lamps once. The formula also covers $S = \varnothing$. The independent checker replays every printed answer and performs coordinate-unbounded breadth-first search through distance 12; the argument above, not that finite check, proves the general rule.

Sources and readiness. Cleary and Taback, *Dead end words in lamplighter groups and other wreath products*, §3.1, pp. 4–6, and §4.1, p. 9, supply the lamplighter and dead-end lineage. Druțu and Kapovich, *Geometric Group Theory*, Exercise 7.82, printed p. 233 (PDF 253), gives the wreath-product distance framework. These small tasks are authored specializations. Final-page targets and guide answers were digitally checked; the prototype is **unpiloted and its physical setup unrehearsed**.